

Q1)

Number of solutions to the equation $x_1 x_2 x_3 x_4 = y_1 y_2 y_3 y_4$, where the variables are allowed to take values 0, 1 and -1 only, is -

Ans) 4353

Q2)

All six digit numbers containing each of the digits 1, 2, 3, 4, 5, 6 exactly once, and not divisible by 5, are arranged in decreasing order. Then sum of two middle digits in 200th number in list is

Ans) 10

Q3)

Number of squares that can be formed in 8×8 chessboard so that each square is surrounded by four square of same dimensions, are

Ans) 45

Q4)

Four digit numbers are formed using the digits 1, 2, 3, 4 (repetition is allowed). The number of such four digit numbers divisible by 11 is-

Ans) 44

Q5)

Consider a sequence of 10 A's and 8 B's placed in a row. By a run we mean single or more letters of the same type placed side by side. For example following is an arrangement consisting of 4 runs of A & 4 runs of B : AAABBABBBBAABAAAABB. The number of arrangements of 10 A's and 8 B's in a row so that there are -

(A) 4 runs of A and 4 runs of B is ${}^9C_3 \cdot {}^7C_3$

(B) 4 runs of A and 5 runs of B is ${}^3C_4 \cdot {}^7C_4$

(C) 3 runs of A and 3 runs of B is ${}^2C_2 \cdot {}^7C_2$

(D) 6 runs of A and 6 runs of B is ${}^4C_5 \cdot {}^7C_5$

Q6)

Number of ways to choose 3 positive integers $a < b < c$ so that $abc = 30030$

Ans) 121

Q7)

How many integers between 2020 and 2400 have four distinct digits arranged in increasing order? (For example, 2347 is one integer.).

Ans) 15

Q8)

How many odd positive 3-digit integers are divisible by 3 but do not contain the digit 3?

Ans) 96

Q9)

If there are N integers between 100 and 999, inclusive, have the property that some permutation of its digits is a multiple of 11 between 100 and 999? For example, both 121 and 211 have this property. Then find the value of $\frac{N-1}{5}$.

Ans) 45

Q10)

The numbers 1,2,3,4,5 are to be arranged in a circle. An arrangement is bad if it is not true that for every n from 1 to 15 one can find a subset of the numbers that appear consecutively on the circle that sum to n . Arrangements that differ only by a rotation or a reflection are considered the same. How many different bad arrangements are there?

Q11)

Sam can only remember 10-digit numbers if the first four digits are either exactly the same as the next four digits of the number or the last four digits of the number. For example, Sam can remember 1234123456 and 3444533444, but not 3344443334. How many 10-digit numbers can Sam remember?

Ans) 1799910

Q12)

How many 5-digit sequences have a digit that appears at least 3 times? (For instance, 03005 and 22922 are examples of such sequences).

Ans) 8560

Q13)

The sum of all 3-digit numbers less than or equal to 500, that are formed without using the digit "1" and they all are multiple of 11, is _____.

Ans) 7744

Q14)

The total number of four digit numbers such that each of the first three digits is divisible by the last digit, is equal to _____.

Ans) 1086

Q15)

The number of 5-digit natural numbers, such that the product of their digits is 36, is

Ans) 180

Q16)

The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is ____

Ans) 42

Q17)

The number of 6 digit number that can be formed using the digits 0, 1, 2, 5, 7 and 9 which are divisible by 11 and no digit is repeated is:

Ans) 60

Q18)

The number of different 5 digit numbers greater than 50000 that can be formed using the digits 0, 1, 2, 3, 4, 5, 6, 7, such that the sum of their first and last digits should not be more than 8, is

Ans) 4607

Q19)

The number of natural numbers, between 212 and 999, such that the sum of their digits is 15, is

Ans) 64

Q20)

The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3, is equal to _____

Ans) 36

Q21)

Five digit numbers are formed using the digits 1, 2, 3, 5, 7 with repetitions and are written in descending order with serial numbers. For example, the number 77777 has serial number 1. Then the serial number of 35337 is _____.

Ans) 1436

Q22)

Let 5 digit numbers be constructed using the digits 0, 2, 3, 4, 7, 9 with repetition allowed, and are arranged in ascending order with serial numbers. Then the serial number of the number 42923 is _____.

Ans) 2997

Q23)

The total number of six digit numbers, formed using the digits 4, 5, 9 only and divisible by 6, is _____.

Ans) 81

Q24)

The number of natural numbers lying between 1012 and 23421 that can be formed using the digits 2, 3, 4, 5, 6 (repetition of digits is not allowed) and divisible by 55 is _____.

Ans) 6

Q25)

The number of 7-digit numbers which are multiples of 11 and are formed using all the digits 1, 2, 3, 4, 5, 7 and 9 is _____.

Ans) 576

Q26)

The total number of three-digit numbers, with one digit repeated exactly two times, is _____.

Ans) 243

Q27)

The total number of 4-digit numbers whose greatest common divisor with 54 is 2, is _____.

Ans) 3000

Q 28) A permutation $(a_1, a_2, a_3, a_4, a_5)$ of $(1, 2, 3, 4, 5)$ is heavy-tailed if $a_1 + a_2 < a_4 + a_5$. What is the number of heavy-tailed permutations?

Ans) 48

Q 29) A **palindrome** is a **whole number** that reads the same forwards and backwards. If one neglects the colon, certain times displayed on a digital watch are palindromes. Three examples are: $\boxed{1 : 01}$, $\boxed{4 : 44}$, and $\boxed{12 : 21}$. How many times during a 12-hour period will be palindromes?

Ans) 57

Q 30) A 10-digit arrangement $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$ is called beautiful if (i) when read left to right, $0, 1, 2, 3, 4$ form an increasing sequence, and $5, 6, 7, 8, 9$ form a decreasing sequence, and (ii) 0 is not the leftmost digit. For example, 9807123654 is a beautiful arrangement. Determine the number of beautiful arrangements.

Ans) 126

Q 31) Find the number of ordered pairs of integers (a, b) such that the sequence
 $3, 4, 5, a, b, 30, 40, 50$
is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.

Ans) 228

Q 32) Find the number of ways to place a digit in each cell of a 2×3 grid so that the sum of the two numbers formed by reading left to right is 999, and the sum of the three numbers formed by reading top to bottom is 99. The grid below is an example of such an arrangement because $8 + 991 = 999$ and $9 + 9 + 81 = 99$.

0	0	8
9	9	1

Ans) 45

Q33) The numbers 1447, 1005 and 1231 have something in common: each is a 4-digit number beginning with 1 that has exactly two identical digits. How many such numbers are there?

Ans) 432